DiarizBench

speaker diarization leaderboard

PILOT · PROTOCOL v1.0

24 recordings · 7 languages · 205 min scored audio
frozen 2026-11-11 · all values from saved runs

Domain **All** | Language **All** | Speakers **Any** | Overlap **Any** | View **Recording mean** | **All modes** | Offline | Streaming

Ranking metric: **DER, zero collar, overlap scored, speaker count unknown**. Offline ranks only. A win requires a Bonferroni-corrected paired-bootstrap difference *and* at least 1 pp. Rows in different operating modes are never ranked against each other.

#	MODEL · VERSION	MODE	DER % · 95% CI	JER %	COUNT	RTF	COVER.	EXPOSURE
1	**CIIL Anuvani** `CIIL/Anuvani-Diarization`	OFFLINE	**13.8** 10.0 to 17.9	29.9 22.7–37.4	33% 8/24 exact	0.053 18.8× real	24/24 complete	corpus exposure Indic; AMI/Vox parent
2	**pyannote community-1** `pyannote/speaker-diarization-community-1`	OFFLINE	**20.4** 15.5 to 25.3	38.6 30.3–47.2	38% 9/24 exact	0.072 13.9× real	24/24 complete	known overlap AMI · VoxConverse
—	**NVIDIA Nemotron-3** `nvidia/Nemotron-3-Diarization`	STREAMING	**21.6** 15.7 to 27.6	32.5 23.4–41.6	58% 14/24 exact	0.079 12.7× real	24/24 complete	unknown no declaration
—	ILLUSTRATIVE API / on-device lane: not evaluated; no invented scores. The broader catalog is in the main report.							

Ranks 1 and 2 apply only to the two offline systems. Nemotron is a separately labelled streaming configuration, with no shared rank. Every measured row is 24/24 complete. Prototype filters are illustrated here; the companion HTML makes them functional.

COMPARISON WITHOUT A FALSE TIE

The community-1/Nemotron gap is **–1.2 pp** (95% CI –4.2 to 2.0). This is not statistically resolved, not proof of equivalence. Anuvani's observed gaps are **6.6** and **7.8 pp**; only the first is a matched offline comparison.

WHAT DER ALONE HIDES

Nemotron has the highest observed count accuracy (14/24 exact) and low-activity recall (61%). Compared with community-1, the gaps are 20.8 pp and 15.8 pp respectively. Coverage differences are descriptive; their superiority was not tested.

BALANCED CELLS · DER % BY CORPUS

SYSTEM	INDIC DIARBENCH (12)	AMI-SDM (6)	VOXCONVERSE (6)	CORPUS-EQUAL
CIIL Anuvani	**16.5**	**17.5**	**4.9**	**12.9**
pyannote community-1	26.1	22.7	6.8	18.5
NVIDIA Nemotron-3	25.1	29.6	6.5	20.4

Best value in each column in green. Anuvani leads all four cells, so the ordering does not depend on corpus weight; the *margin* does (8.6–9.6 pp on Indic against 1.6–1.9 pp on VoxConverse). The 24-recording average stays labelled exploratory because the references and conversation content differ; their separate effects were not measured.

> **Reading rule.** Per-corpus cells come first. The table uses a recording mean; the secondary view equal-weights three corpus means. Eight balanced domain × language cells are a production design, not measured here.

24	7	3	72/72	196	0
recordings	languages	systems	units completed	min reference speech	errored runs

Model detail

one row, opened

MEASURED

protocol v1.0 · run 2026-11-10 · 24/24 complete

CIIL Anuvani

CIIL/Anuvani-Diarization

Compare with **all systems** ▾ Slice **All corpora** ▾

ERROR DECOMPOSITION — WHERE THE 13.8% GOES



missed speech · false alarm · confusion — sums to DER by construction

The only balanced profile in the pilot. community-1 and Nemotron lose most of their error to missed speech (10.6 and 17.1 pp), which points at speech-activity behavior; the cause was not isolated.

RUN CONDITIONS

Family	**end-to-end**
Operating mode	**offline**
Hardware	**Tesla T4 ×2**
Settings	**documented defaults**
Speaker count	**unknown to system**
Licence	**CC BY-NC 4.0**
Exposure	**Indic + parent AMI/Vox**

SPEAKER COVERAGE

Exact count	**8/24 = 33%**
Count MAE	**0.83 speakers**
Count bias	**–0.58 speakers**
Low-activity recall	**47% (18/38)**
95% CI (clustered)	**14 to 72%**
At 3% / 10% activity	**42% / 53%**

Under-counts on average and recovers under half of the speakers who talk least. Stable across the sensitivity grid, so the finding is not a threshold artefact.

PER CORPUS — NEVER POOLED SILENTLY

Indic DiarBench (12)	**16.5%**
AMI-SDM (6)	**17.5%**
VoxConverse (6)	**4.9%**
corpus-equal	**12.9%**

Leads all three corpora, so the ranking does not depend on corpus weight. The margin does: 8.6–9.6 pp on Indic, 1.6–1.9 pp on VoxConverse.

SPEED, AS MEASURED

RTF (panel mean)	**0.053**
Speed factor	**18.8× real time**
Setup/load	**12.98 s**
5-min projection (setup + RTF)	**29.0 s**
Break-even vs community-1	**about 11 min audio**

Projected totals change with file length and setup reuse. Timed callables differ in I/O and post-processing; these are implementation timings, not matched service latency.

HOW THIS ROW COMPARES

	ANUVANI	COMMUNITY-1	NEMOTRON-3
DER %	**13.8**	**20.4**	**21.6**
JER %	**29.9**	**38.6**	**32.5**
Exact count	**8/24**	**9/24**	**14/24**
Low-activity recall	**47%**	**45%**	**61%**
RTF	**0.053**	**0.072**	**0.079**

EXAMPLE FAILURE — THE HONEST ONE

BN-01, Bengali near-field. Reference speaker SPEAKER_05 contributes 12.87 s in six interjections across 168 s. All three systems drop the speaker entirely, Anuvani included. RTTM alignment suggests merging; no audio listening audit was performed. In this file, DER stays at 22.8% while a participant disappears from the transcript.

TE-02, Telugu in-the-wild. Three simultaneous speakers at 97.2–97.7 s. Local label counts do not establish correct speaker identities; overlap-only DER on this file is 91.7%. Overlap, not clean single-speaker speech, is what still separates these systems — and no system in the pilot handles it well.

What this row does not tell you. Anuvani was fine-tuned on Indic DiarBench, one of the three pilot corpora. Its Indic margin cannot be separated from training-data exposure, and the exposure chip stays on every surface where the number appears.

Governance

how the board stays comparable

OPERATING RULES

protocol v1.0 · references v1 · immutable once published

HOW A SYSTEM GETS ON THE BOARD

CURATED TRACK — RANKED

- **Pinned checkpoint.** Exact repository ID and revision, not a family name.
- **Documented defaults.** Settings fixed before any pilot audio is scored; no per-corpus tuning.
- **Exposure declaration.** Known overlap, declared disjoint, or unknown — printed on the row.
- **Complete run.** Every recording in the frozen manifest, or the row is published as partial with coverage shown and no rank.
- **Independent reproduction.** Maintainer re-runs from the pinned environment before the row is marked verified.

BRING-YOUR-OWN TRACK — SEPARATE, UNRANKED

- Submitters run the harness themselves and publish hypotheses plus a run log.
- Self-reported settings are accepted but labelled **self-reported**.
- No rank and no head-to-head claim until the maintainer reproduces the run.
- Never merged into the curated table, even after reproduction — it moves track, keeping the provenance of every number legible.
- Closed systems that cannot be run locally stay in the API / on-device lanes, where timing is not comparable to self-hosted RTF.

SUBMISSION LIFECYCLE

01 Submit — Checkpoint ID, licence, exposure statement, run log. ›

02 Triage — Access, licence, and whether the system is runnable at all. ›

03 Run — Frozen manifest, pinned environment, checkpointed driver. ›

04 Score — Contract scorer plus the fixture suite; artefacts archived. ›

05 Review — Failure cases read by a human; anomalies published, not hidden. ›

06 Publish — Row carries protocol version, reference version, run date, artefact hash.

RE-SCORING WHEN A REFERENCE IS CORRECTED

Reference corrections are the normal case, not an exception: a mislabelled speaker or a bad turn boundary changes every system's score at once. The rule is **re-score everything, publish everything**.

- **Version the reference, never edit it.** A correction creates `references v2`; v1 stays retrievable.
- **Full re-score.** Every system is scored against v2 with the same protocol. No partial refresh, no cherry-picked rows.
- **Publish the diff.** Old score, new score, and the delta per system, side by side.
- **Bump the protocol version** only when a setting changes. Reference corrections and setting changes are logged separately, because they mean different things.

VERSIONING, ARCHIVAL, AND CADENCE

EVERY PUBLISHED ROW CARRIES

Protocol version	**v1.0**
Reference version	**v1**
Checkpoint revision	**pinned commit**
Run date and hardware	**2026-11-10 · T4 ×2**
Artefact hash	**hypotheses + results**

CADENCE

Full refresh	**quarterly**
New systems	**monthly intake**
Out-of-cycle	**reference correction**
Old leaderboards	**archived read-only**
Silent re-rank	**never**

WHAT A CHANGE LOOKS LIKE — EXAMPLE LOG

DATE	PROTOCOL	REFERENCES	CHANGE	WHAT IS PUBLISHED
2026-11-11	v1.0	v1	Initial pilot, 3 systems	All rows, protocol v1.0
2027-02-01	v1.0	**v2**	Speaker split fixed on 3 files	Full re-score, old / new / delta
2027-03-15	**v1.1**	v2	Collar sensitivity added to the contract	Version bump, all rows re-scored

Deliberately out of scope. cpWER/tcpWER and WER/CER are future speaker-attributed ASR metrics, requiring transcripts and timing. Add diverse datasets (DIHARD, ICSI, CALLHOME, DISPLACE, AliMeeting, CHiME-8) and live latency as separate tracks. None was evaluated in this pilot.

Audio inspector hear the evidence

REAL PILOT EXCERPT

Functional playback in companion HTML

TE-02 · TELUGU · REFERENCE OVERLAP EXAMPLE

Original source interval 92–104 s. Three reference speakers overlap at 97.2–97.7 s. Inspect the saved turns and listen before interpreting an aggregate score.

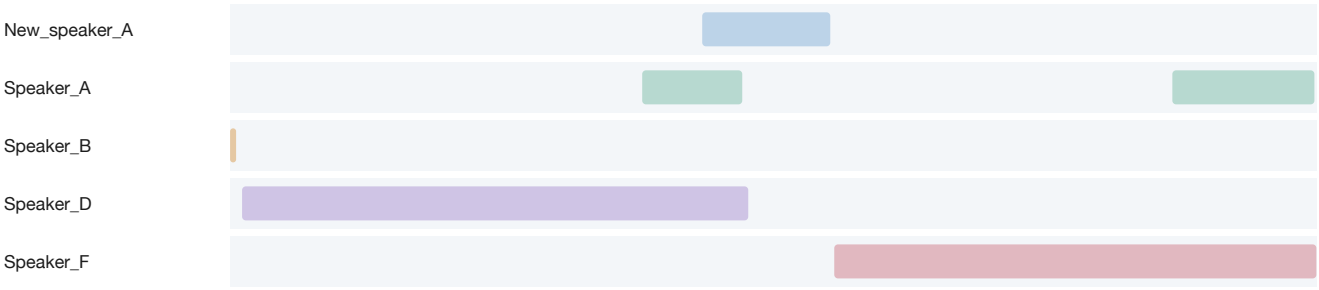
Example **TE-02 · Telugu** ▾ | System **Anuvani · offline** ▾ | Speed **1×** ▾ | + Open local audio

▷ **Original mixture** 0:00 / 0:12

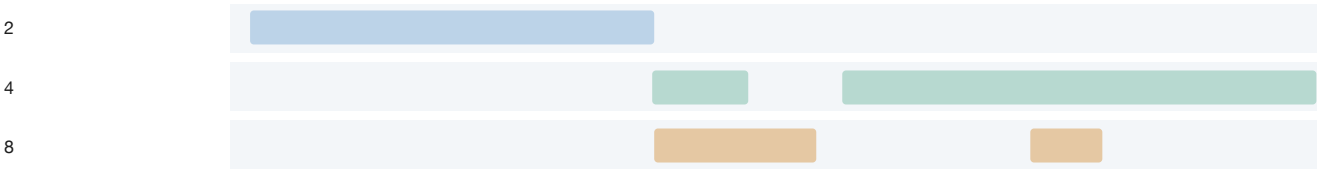


92 s source 98 s 104 s

REFERENCE · NATIVE CORPUS LABELS



ANUVANI · SAVED HYPOTHESIS



Colors and speaker IDs are local to each track. Same colors do not establish cross-track identity matching.

LISTENING CONTROLS

Play/pause, seek the waveform, set playback speed, zoom the timeline, and click any reference/model turn. Speaker previews play the original source interval, including other voices during overlap.

WHAT IS MEASURED

Waveform: actual audio amplitude. Turns: saved reference and hypothesis RTTMs. Full-recording DER stays labelled separately from this short excerpt. No separated stems or computed similarity scores are presented.

How to try it. Open **Ayush_Kumar_Q1_Leaderboard.html**, then Audio inspector. Bengali, Telugu and Tamil examples are embedded, so playback requires no network. A reviewer can also choose a local file for unscored playback.

Audio: Sarvam AI, Indic DiarBench, telugu_itw_004 (use the companion HTML for exact source/sample attribution), CC BY 4.0. Excerpted and converted to mono 16 kHz PCM. Dataset: https://huggingface.co/datasets/sarvamai/indic-diarbench